AI vs. Human: Data-Driven Prediction Model for Brain Stroke

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ABSTRACT

Brain stroke is a time sensitive medical condition caused by a blockage or rupture of blood flow to the brain, and patient outcomes depend heavily on rapid identification and treatment. This study investigates which predictors are strongly associated with brain stroke prevalence and evaluates stroke prediction abilities of statistical and machine learning models. Using a brain stroke patient dataset with 5% stroke rate, I built logistic regression, decision tree, and random forest models to analyze predictor importance. These analyses were conducted through two workflows: human-computed statistical analysis and AI-driven data analysis. Comparing these approaches allowed the study to examine whether AI-assisted analysis can reproduce or complement traditional statistical workflows. Overall, the findings suggest that statistical and machine learning models may support stroke risk screening when interpreted alongside clinical insight. The workflow comparisons also provides insights to potential strengths and limitations of AI.

METHODS

Data Cleaning & Processing

A brain stroke patient dataset containing patient demographic, clinical, and stroke-indicating characteristics was obtained from Kaggle, an open data source. The original data was quality-controlled and processed for any missing entries prior to analysis.

Data Visualization & Exploratory Analysis

Data visualizations interactive dashboards were built using Tableau to examine variable relationships, patterns, and descriptive statistics

Train & Test Data Split

The cleaned original dataset was split up into 50% training and 50% testing dataset using random distribution while keeping the stroke prevalence of each dataset even and consistent.

Statistical & Machine Learning Analysis

I) Logistic regression II) Decision tree, and III) Random forest, were built using the training set and evaluated with the testing set. Results from each model were interpretated and compared to draw meaningful conclusions on influential predictors of stroke. The models were built using two workflows mapped below:

1) Human-driven Analysis Workflow

1) Build & fit initial model using training data → 2) Evaluate using testing data (sensitivity, specificity) → 3) Interpret results & Draw conclusions

2) AI-driven Analysis Workflow

1) Prompter: Generate structured prompt → 2) Analyst: Build model & produce results/outputs → 3) Recorder: report results/interpretations

RESULTS

I. Logistic Regression

• Significant Predictors

- Age is the most significant predictor selected by both models (p-values <0.001)
- Top three predictors are consistent

Human-driven Model	
Rank	Predictor
1	Age
2	Hypertension
3	Average Glucose Level

AI-driven Model	
Rank	Predictor
1	Age
2	Average Glucose Level
3	Hypertension

• Test-set Evaluation

	Human-driven Model	AI-driven Model
Accuracy	0.7064	0.6929
Sensitivity	0.8560	0.8468
Specificity	0.6985	0.6848

- Both models show similar performances in terms of stroke prediction.
- Slightly higher metrics in human-driven model suggests improved screening capabilities

II. Decision Tree

• Significant Predictors

Human-driven Model	
Rank	Predictor
1	Age
2	Average Glucose Level
3	BMI

AI-driven Model	
Rank	Predictor
1	Age
2	Gender (Male)
3	Average Glucose Level

- Both models identified age as the most model-associated predictor
- AI-driven model only used age as the only splitting variable → all other predictors have zero importance
- Overall, the human-driven model provides more detailed information about predictor relationships

• Test-set Evaluation

	Human-driven Model	AI-driven Model
Accuracy	0.6916	0.6736
Sensitivity	0.8240	0.8306
Specificity	0.6846	0.6654

- Tradeoff in sensitivity and specificity
- AI-driven model identifies stroke cases better while human-driven model identifies non-stroke cases better

III. Random Forest

• Significant Predictors

- Different leading predictor
- Consistent set of predictors differently ordered. This suggests that age, average glucose level and BMI may have clinical associations to stroke.

Human-driven Model	
Rank	Predictor
1	Age
2	Average Glucose Level
3	BMI

AI-driven Model	
Rank	Predictor
1	Average Glucose Level
2	Age
3	BMI

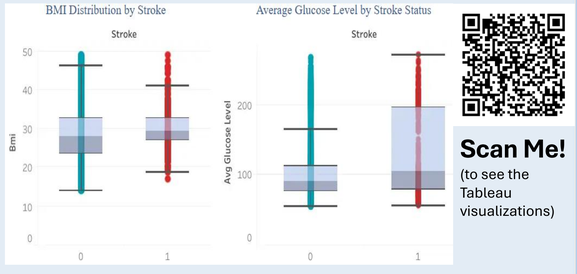
• Test-set Evaluation

	Human-driven Model	AI-driven Model
Accuracy	0.7020	0.7110
Sensitivity	0.8080	0.8145
Specificity	0.6964	0.7055

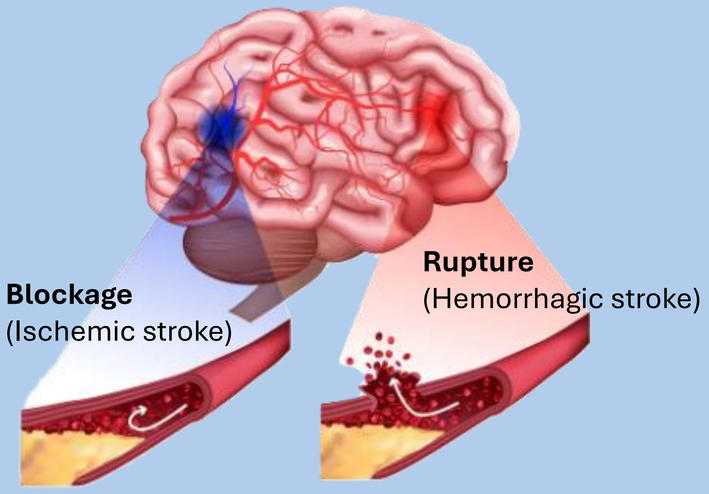
- Slightly higher metrics for the AI-driven analysis
- Higher sensitivity → AI-driven model misses fewer stroke cases
- Higher specificity → AI-driven model creates fewer false positives

IV. Data Visualizations





Scan Me!
(to see the Tableau visualizations)



COMPARISON

I. Analysis Time & Efficiency

- Overall, the AI-driven workflow consumed less amount of time with auto-generative abilities
- Tradeoff: Despite the speed and convenience, critical errors were made in the AI-driven analysis, which drops reliability and questions its true efficiency

	Human-driven	AI-driven
Average analysis time	3-4 hours	0.5-1 hour

II. Error in AI-driven Analysis

	Best Threshold Value	
	Human-driven	AI-driven
Log. Reg.	0.0425	0.4478
Decision Tree	0.0512	0.6094
Random Forest	0.0721	0.0420

- Misinterpretation of ROC curve:
  - False positive rate (x=0.4478) treated as threshold value.
  - Ex. AI-driven logistic regression model's ROC curve



CONCLUSION

- Overall, the human-driven workflow showed higher interpretability and consistency across model results.
- Based on this study, AI appears most useful as an assistive tool for code generation, workflow organization, and documentation, rather than as a fully independent analytical system.
- The AI could generate working code efficiently, but it still required human oversight to confirm that the methods were applied and interpreted correctly.

ACKNOWLEDGEMENTS

I thank Dr. Wanchunzi Yu for the provision of statistical insight, methodical guidance, research sources, and overall support throughout this project.